Tai Dang

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Education & Training

Stanford University *Visiting Researcher*
Advisors: Thang Luong (Deepmind) & Jeff Glenn, and Sanmi Koyejo (Stanford)

University of Massachusetts Amherst *B.S. '24 in Computer Science*
Advisors: Bruno Castro da Silva (Computer Science)

Research

My research focuses on generative modeling and decision-making, particularly diffusion models and reinforcement learning, with applications in computational biology. My broader goal is to develop ML systems that accelerate scientific discovery and enable advances in biology and medicine. Below are selected papers:

NeurIPS'26 1. **Dang** et al. Plausible Biomolecular Structure Prediction via Physics-informed Reinforcement Learning.
- Achieved SOTA in structure prediction fidelity and affinity via RL with physics-based rewards.
- *Paper:* arXiv, *Blog:* π-Fold, *Code:* nti/pifold

ICML WS'25 2. **Dang** et al. Preferential Multi-Objective Bayesian Optimization for Drug Discovery.
- Recovered 16/37 EGFR and 37/58 DRD2 drugs from 100K ligands via chemist-guided screening.
- *Paper:* arXiv, *Code:* tai-dang11/cheapvs

EACL'23 3. **Dang** et al. Enriching Biomedical Knowledge for Low-resource Language Through Translation.
- SOTA in Vietnamese biomedical benchmark and high-quality Vietnamese MedNLI dataset.
- *Paper:* eacl, *Code:* vietai/ViPubmed

Preprint'22 4. MTet: Multi-domain Translation for English and Vietnamese.
- *Details:* SOTA in English-Vietnamese translation and high-quality multi-domain bilingual corpus.
- *Paper:* arXiv, *Code:* vietai/mTet

COLING'25 5. AURORA-M: Open Source Continual Pre-training for Multilingual Language and Code.
- Developed a 15B open-source multilingual model continually pre-trained on code and text.
- *Paper:* coling, *Model:* huggingface.co/aurora-m

ICML WS'26 6. Gathering Context that Supports Decisions via Entropy Search with Language Models.
- Closed 85% of performance gap to fully-informed agents via uncertainty-driven information seeking.

Experience

1. Xaira Therapeutics – *AI Resident '26–Present*
   - De Novo Design Team
2. Stanford University – *Visiting Researcher '26*
   - Post-trained AlphaFold 3 via reinforcement learning, SOTA on structure fidelity.
   - Achieved optimal drug screening on large libraries via Bayesian optimization.
3. UMass Amherst – *Research Assistant '23*
   - Engineered a multi-modal retrieval system for VQA task, boosting retrieval accuracy by 5%.
   - Enhanced policy specialization by optimizing discount factors within the OLS Convex Coverage Set.
4. Ontocord – *Research Intern '23*
   - Distilled 7B LLM model to **5x** smaller size while maintaining performance parity.
   - Developed open-source Vietnamese LLM by processing 1TB of data.
5. EOG Resources – *Software Engineer Intern '23*
   - Built a graph-based visualization tool that accelerated complex data analysis workflows by 10%.
   - Automated CI/CD pipelines and enhanced security by migrating repositories to GitHub Actions.
6. VietAI – *Research Intern '22*
   - Developed SOTA En-Vi model and improved Biomedical NMT via self-training.
   - Drove 6% BLEU improvement in Biomedical NMT via self-training and released Vi-MedNLI dataset.
7. FPT Software – *Research Intern '21*
   - Developed an interactive AI frontend, streamlining workflows for 50+ engineers.

Projects

1. Multi-Objective GFlowNet for Drug Design
   - Generated diverse, synthesizable molecules with optimized affinity and ADMET via SynFlowNet.

Honors & Funding

Paper Awards: ICLR'25 Workshop Spotlight, ICML'25 Workshop Spotlight
Grants: Google-HAI Grant '24 ($90,000), Google-HAI Grant '25 ($100,000), Stanford Marlowe Grant.

Skills

Frameworks: PyTorch, Flax, JAX, TensorFlow, Hugging Face, Flask, Node.js, Neo4j
Languages: Python, Java, JavaScript, C/C++, SQL
Tools: Git, Linux, GCP, Slurm, Docker, GitHub, Kubernetes